RV College of Engineering®
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Department of Biotechnology

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Academic year 2025-2026 (Odd Sem)

Date	December, 2025	Maximum Marks	10 + 50
Course Code	BT232TC	Duration	120 min
Sem	III SEMESTER	Closed Book Offline Quiz and Test 2	

BIOSAFETY STANDARDS AND ETHICS (BASKET COURSE)

Answer all questions in Part A & Part B. Answer Part A in first two pages in sequence only

Part A

Sl No	Questions	M	BT-L	CO
1	List out the types of licenses of FSSAI.	1	1	3
2	Name two water borne diseases and responsible organisms	1	1	3
3	How do you identify the food spoilage?	1	2	3
4	Expand GLP,GAP,FSSAI and HACCP	1	1	3
5	What is 2hr- 4hr rule for food storage	1	2	4
6	Define blanching method of food processing	1	2	4
7	In FSSAI code what is the 4th digit represent	1	3	3
8	After sterilization of canned foods, what will be the shelf life?	1	1	4
9	Define allergens give examples	1	1	3
10	What are food preservatives? give examples	1	1	

Part B

Sl. No	Questions (Test)	M	BT -L	CO
1	The various steps involved during the processing of raw rice is mentioned below. Please mention type of hazard and how to prevent hazards in each step **Pre-Cleaning**: Paddy is cleaned to remove straw, dust, stones, and other impurities using screens and gravity separators to protect machinery and improve quality. **De-Stoning**: Specific machines separate heavy impurities like stones from the grain. **Dehulling (Husking)**: A sheller removes the inedible husk from the paddy, creating brown rice (bran layer intact). **Paddy Separation**: Unhusked paddy is separated from the brown rice for re-milling, ensuring uniformity. **Whitening (Milling)**: Brown rice is milled to gradually remove the bran layer and germ, producing white rice. **Polishing**: An optional step that adds shine and removes final traces of bran for better appearance. **Grading & Sorting**: Rice is separated by size (head rice, broken rice) and quality, often using color sorters. **Packaging**: Final rice is weighed and packed for storage and sale.	10	3	3
2	Elaborate the principles of HACCP in detail	10	3	4
3	Illustrate the GMP and GLP followed in food industries	10	3	3
4	Explain the various testing methods and equipments used in food testing laboratories	10	2	3
5	Discuss in detail the packaging technology. add a note on innovative packing methods	10	2	3

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NBA Accredited (UG - 6Years)

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Department of Biotechnology

DEPARTMENT OF BIOTECHNOLOGY

Date	Jan 14th , 2026	Maximum Marks	50+10
Course Code	BT232TC	Duration	120 Min
Sem	3rd	Improvement test CIE-3	
BIOSAFETY STANDARDS AND ETHICS (Basket Course)			

Instructions: Answer all questions

S. No	Quiz Questions	MKS	CO	BTL
1	Why is blanching done before freezing vegetables?	1	3	2
2	Which principle of food preservation is involved in drying?	1	3	2
3	Name one hazard controlled by HACCP in food processing.	1	4	1
4	Name one food safety risk associated with the use of antibiotics in animals.	1	4	1
5	How do food additives improve food safety? Give one example.	1	3	2
6	Differentiate between GMP and GAP with any one example.	2	3	3
7	Why biodegradable packaging materials are considered novel and important?	2	3	3
8	Differentiate between biosafety and bioethics.	2	4	3

S. No	Test Questions	MKS	CO	BTL
1	Mention the various types of food allergens and their impact on human health.	10	3	3
2	Illustrate the various stages where hormones and antibiotics will be given for the animals, and mentions their effect on human beings.	10	4	4
3	Explain the importance of Research Ethics and Clinical Ethics in life	10	4	4
4	List out various Food Preservatives available in India and their effect on health	10	3	3
5	Elaborate on various parameters considered for the basic price fixation on food products	10	3	3

2b	Explain the major differences in terms of air recirculation between various types of Class II Biosafety Cabinets.Draw a neat diagrm of class 2 biosafety cabinets.	08	2	4
3a	Farmers in the Northern part of India are protesting against the genetically modified Mustard seeds. In this context discuss in detail the pros and cons of the GM Mustard seeds.	16	2	3
	OR			
4a	Discuss in detail the Biosafety Guidelines of the Government of India and their alignment with international standards.	08	2	3
4b	Evaluate the role of Institutional Biosafety Committees (IBCs) and the Review Committee on Genetic Manipulation (RCGM) as per Indian biosafety guidelines.	08	2	5
5a	Elaborate on methods of food analysis add a note on the instruments used for analysis	08	3	3
5b	Elaborate on the sources and types of microbial contamination in food and the preventive strategies employed to control them.	08	3	5
	OR			
6a	Why food testing is considered important ? discuss the parameters and who will be the beneficiary of food testing.	08	3	4
6b	Give an account on industrial importance of beneficial microorganisms in food processing, with examples of organisms and end products.	08	3	3
7a	Discuss the different types of active and intelligent packaging methods and their applications.	08	3	4
7b	Elaborate on smart packaging and its importance in detail.	08	3	5
	OR			
8a	What are Good Production Practices (GPP) in the food industry? Discuss their importance.	08	3	3
8b	What are novel packaging materials, and how do they differ from traditional packaging materials?	08	3	3
9a	Elaborate on case study of Food Allergens, examples,symptoms,and treatment.	08	4	3
9b	What is plagiarism and which all activities were considered as plagiarism?	08	4	3
	OR			
10a	Elaborate on the the role of food preservation in food safety for the benefit of consumers	08	4	5
10b	Mention the factors which contributes to food borne illness? Elaborate on each of the factors.	08	4	4

RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

III Semester B.E. Regular / Supplementary Examination February/March 2026

Bio Safety Standards and Ethics

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	List the roles of the legal and regulatory framework for biosafety levels in India.	02	2	1
1.2	Define GAP and how it helps in food production?	02	3	1
1.3	Give the role of different types of microorganisms in food spoilage	02	3	2
1.4	Give some examples of chemical presevatives and uses	02	4	1
1.5	Define Plagiarism and mention its implications	02	4	2
1.6	What is the primary purpose of a Biosafety Cabinet (BSC)?	02	2	1
1.7	List the roles of the legal and regulatory framework for biosafety levels in India.	02	2	1
1.8	State two general principles of food microbiology.	02	3	2
1.9	Give examples and advantages of smart or intelligent packing of food.	02	4	3
1.10	Give reasons for labeling of food additives and allergens.	02	3	2

Part B

Question No	Question	M	CO	BT
2a	Explain the construction, classification, working, and applications of Biosafety Cabinets in biotechnology and biomedical research.	08	2	3

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DEPARTMENT OF
BIOTECHNOLOGY
Academic year 2025-2026 (Odd Sem)

Date	6th November 2025	Maximum Marks	60
Course Code	BT242TC	Duration	120 min
Sem	III Semester	Closed Book Offline Quiz & Test -1	

Biosafety Standards and Ethics (Basket course)

Sl. No	Questions (Quiz)	M	BT-L	CO
1.	Define containment give example	1	2	1
2.	A street dog bites a person, a person is infected and this comes to which level of Biosafety?	1	2	1
3.	Mention the mechanism involved in HEPA filter	1	1	1
4.	PPE with air pressure is considered to be which BSL?	1	2	2
5.	Amoebic meningoencephalitis cases reported in Kerala, the infected person condition reaches to which biosafety level.	1	2	2
6.	The Cartagena Protocol was adopted in which year in India.	1	1	1
7.	Expand the term ULPA.	1	2	1
8.	What is the composition of IBSC.	1	2	2
9.	Differentiate between Laminar Air Flow and Biosafety cabinet	2	1	1

Sl. No	Questions (Test)	M	BT-L	CO
1.	Classify various types of Biohazards, and their impact on human health and environment.	10	2	1
2.	Compare Biosafety level class II cabinet with Biosafety level class III cabinet with a neat diagram	10	3	1
3.	India not signed an agreement with US for genetically modified seeds. In this context discuss in detail the pros and cons of the GM.	10	3	2
4.	Explain the different generation of GM crops with examples.	10	3	2
5.	Explains the function of Advisory committee, Approval committee and Monitoring committee responsible for GMOs in India.	10	2	2

BT-L-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Test Max Marks	27	33	---	---	4	26	30	---	---	---

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